

Mohammed Shan P S

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of professional experience building full-stack web applications and backend services using JavaScript, TypeScript, React, Node.js, Express.js, and MongoDB. Experienced in building RESTful APIs, database-backed applications, and responsive user interfaces, with enterprise software engineering experience at Visa and client-facing development experience across diverse projects. MSc Cyber Security (Distinction) with practical knowledge of secure software development, authentication and authorisation, and OWASP-aligned security principles. Eligible to work in the UK without sponsorship and open to relocation across the UK.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL, Python, Java, C++

Frontend: React.js, Redux, HTML5, CSS3, Tailwind CSS, Responsive Web Design, Vite

Backend: Node.js, Express.js, REST APIs, JWT, Mongoose, WebSockets

Databases: MongoDB, PostgreSQL, MySQL, Redis, Prisma, Database Design

Cloud & DevOps: AWS, Docker, Git, GitHub, Linux, npm, CI/CD

Testing: Jest, Unit Testing, Integration Testing

Security: Secure Coding Practices, OWASP Top 10, Authentication & Authorisation, API Security

AI-Assisted Development: GitHub Copilot, Claude

WORK EXPERIENCE

Software Engineer | Freelance

Remote (India) | Sep 2023 – Nov 2024

- Developed full-stack web applications and backend services using React, TypeScript, Node.js, Express.js, and MongoDB, delivering production-ready solutions tailored to client requirements
- Engineered RESTful APIs, authentication and authorisation mechanisms, and database-driven application functionality to support business workflows
- Built responsive, component-based user interfaces using React and modern frontend development practices to improve usability across devices
- Collaborated with clients to gather requirements, translate business needs into technical solutions, and deliver features throughout the software development lifecycle

Software Engineer | Visa

Bangalore, India | Jun 2022 – Aug 2023

- Remediated security vulnerabilities identified through code reviews and security scanning tools in Java backend services, applying secure coding practices
- Architected and built an internal vulnerability management dashboard using React, Node.js, Express.js, and MongoDB, consolidating findings from multiple security tools and improving visibility across engineering teams
- Contributed to a large-scale data centre migration by implementing service and configuration changes across production systems while maintaining reliability and operational continuity
- Collaborated with cross-functional teams to support deployment, validation, and integration activities across distributed enterprise systems

Software Engineer Intern | Visa

Bangalore, India | May 2021 – Jul 2021

- Created a Testcontainers-based integration testing solution to migrate test environments from an in-memory H2 database to containerised MySQL
- Improved integration test reliability by enabling database-backed testing workflows aligned with production system behaviour

Software Validation Engineer | Broadcom

Bangalore, India | Jun 2018 – Nov 2018

- Developed automated WLAN test execution scripts using Python within an internal test automation framework, reducing manual testing effort and enabling unattended execution

PROJECTS

Real-Time Chat Application | Node.js, React, Express.js, MongoDB, Socket.IO | [GitHub](#)

- Designed and developed a real-time messaging platform using Socket.IO, supporting low-latency communication between concurrent users
- Implemented a full-stack, event-driven architecture combining WebSockets and REST APIs with a responsive React-based user interface

Swift Typer Application | React, Tailwind CSS | [GitHub](#) | [Live Demo](#)

- Built an interactive typing speed application with real-time performance tracking and theme-switching functionality
- Implemented a responsive, component-based UI with smooth animations to enhance user experience

Dynamic Graph Algorithms for GPU | CUDA, C++ | [GitHub](#)

- Engineered GPU-accelerated algorithms for dynamic graph processing as part of the Starplat DSL research project, enhancing computational efficiency for large-scale graph operations

EDUCATION

MSc Cyber Security

University of Chester, United Kingdom | Jan 2025 – Feb 2026 | Distinction

MTech Computer Science and Engineering

Indian Institute of Technology Madras, India | Aug 2020 – Jun 2022 | CGPA: 9.44

BTech Computer Science and Engineering

Cochin University of Science and Technology, India | Aug 2014 – Jun 2018 | CGPA: 7.47

ACHIEVEMENTS

- **GATE CSE Rank 591** – Secured All India Rank 591 among approximately 100,000 candidates
- **Prakash Arora Prize**, IIT Madras – Awarded for highest CGPA in first year of MTech
- **Star Teaching Assistant**, IIT Madras – Recognised for exceptional academic support in Computer Science courses
- **Graph DSL Patent Contributor**, IIT Madras – Contributed to the development of a patented Domain-Specific Language for graph processing
- **Student Voice Representative**, University of Chester – Represented student feedback to faculty, contributing to academic and welfare improvements